DATA & ANALYTICS WEDNESDAY · SYDNEY

Science in the era of cognitive abundance

Tobias J. Osborne · Leibniz Universität Hannover

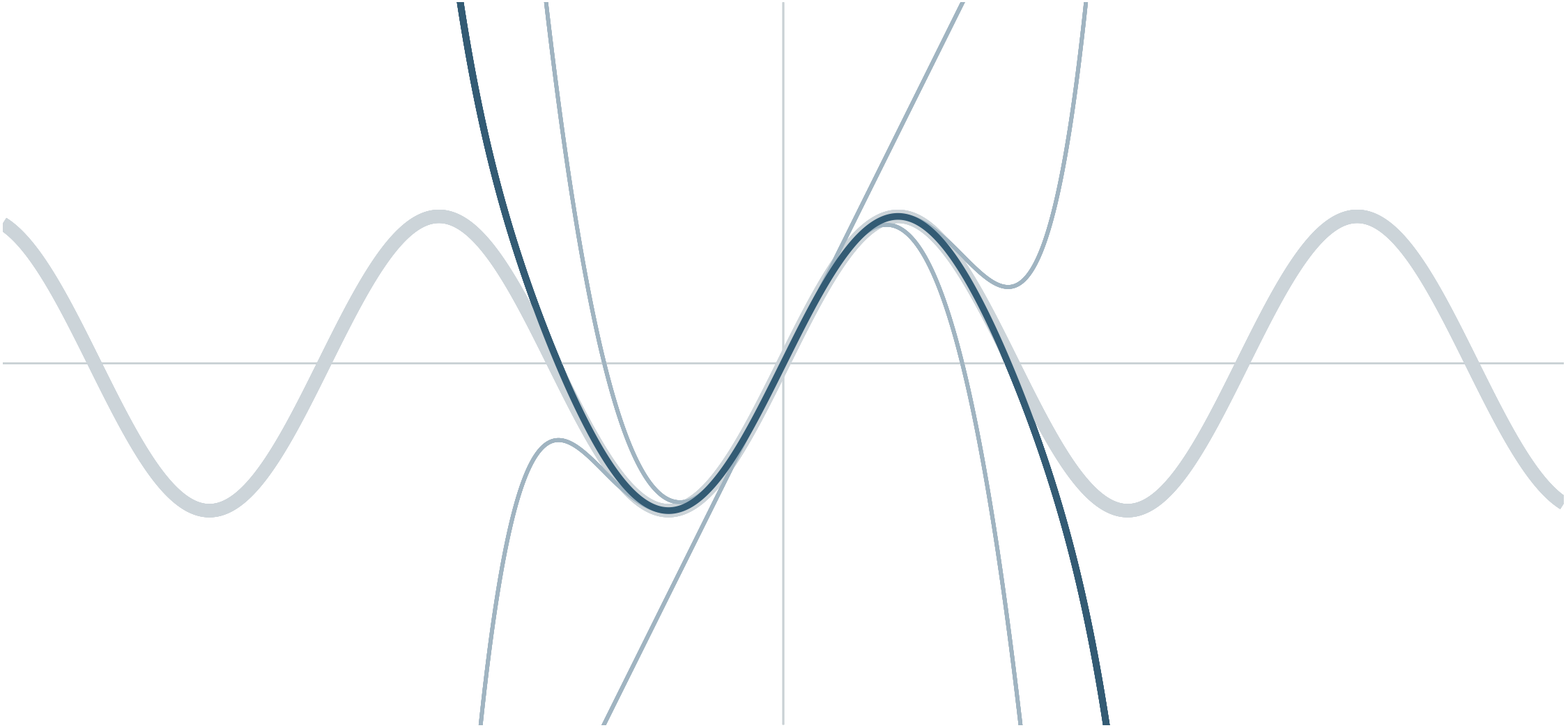
14 October 2026

$$\sin x \approx x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!}$$



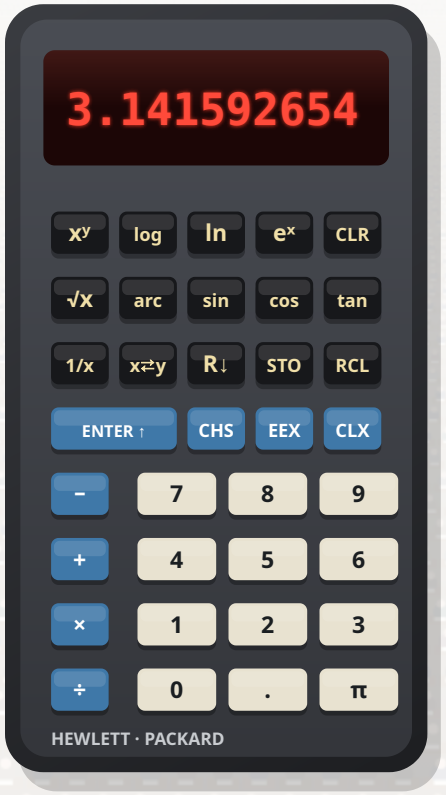
Colin Maclaurin
1698 to 1746

4
terms, by hand



64,000 computers

L. F. Richardson’s forecast factory, imagined in 1922



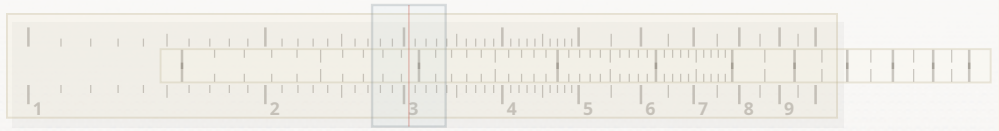
HP-35

January 1972

$395

100,000 sold in 1972

the sales goal was 10,000



1976 TI-30

under $25

1,700,000,000,000

times the performance of computing by hand, at least

Richardson (1922), quoted by P. Lynch (1993) · Computer History Museum · The Henry Ford · Nordhaus (2007), lower end of his range

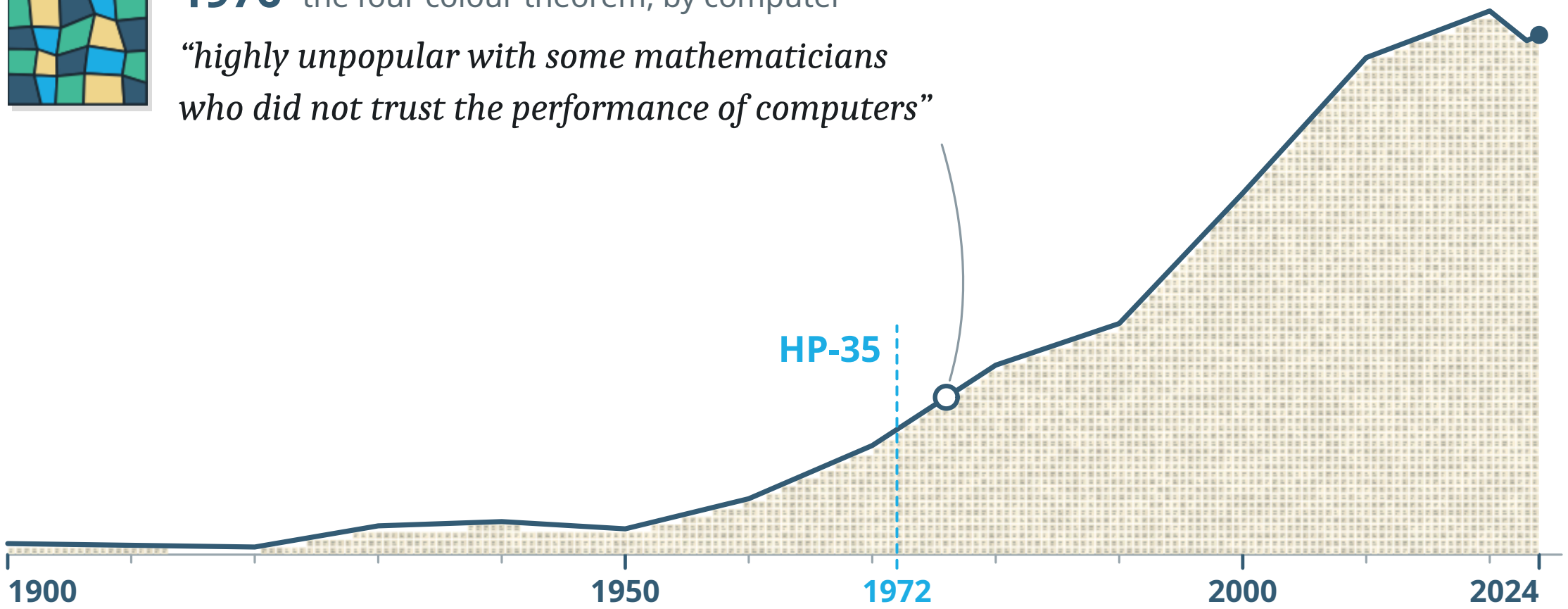
124,917

mathematical publications in the year 2024

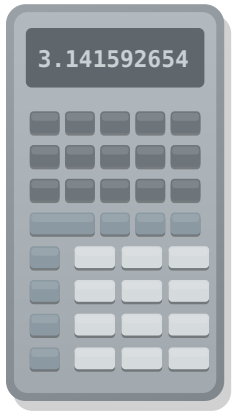


1976 the four colour theorem, by computer

“highly unpopular with some mathematicians who did not trust the performance of computers”



zbMATH Open, records by publication year, read 12 Oct 2026; before 1940 from the *Jahrbuch*, coverage uneven · quotation: *The Boston Globe*, 30 Apr 2013 · tiles are texture



HP-35
January 1972
$395

6 October 2026

719 manuscripts,
written by a machine



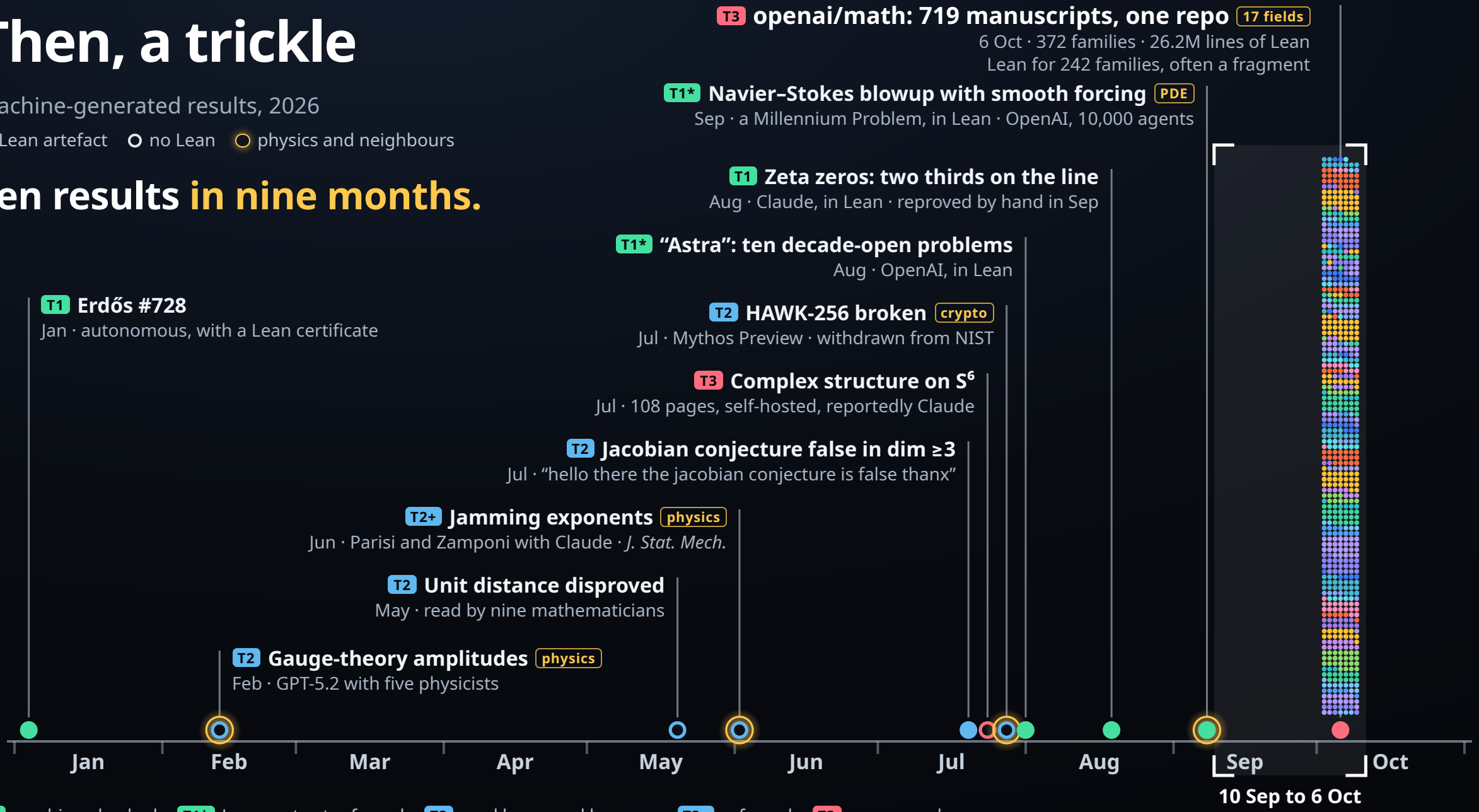
github.com/openai/math: first commit 6 Oct 2026; 722 manuscripts then, 719 after three withdrawals · HP-35: Computer History Museum, HP

Then, a trickle

machine-generated results, 2026

● Lean artefact ○ no Lean ○ physics and neighbours

Ten results in nine months.



T1 machine-checked T1* Lean, not yet refereed T2 read by named humans T2+ refereed T3 announced

dates of first announcement · arXiv, *J. Stat. Mech.*, lab and press announcements, github.com/openai/math (read 12 Oct 2026)

The trickle became a flood

github.com/openai/math · one dot per manuscript, by first folder date

719
manuscripts

372
result families

34,669
PDF pages

26.2M
lines of Lean

about 4,000 problems posed · 3 h of compute per result



52%
in two days,
23 and 24 Sep

194
177
112

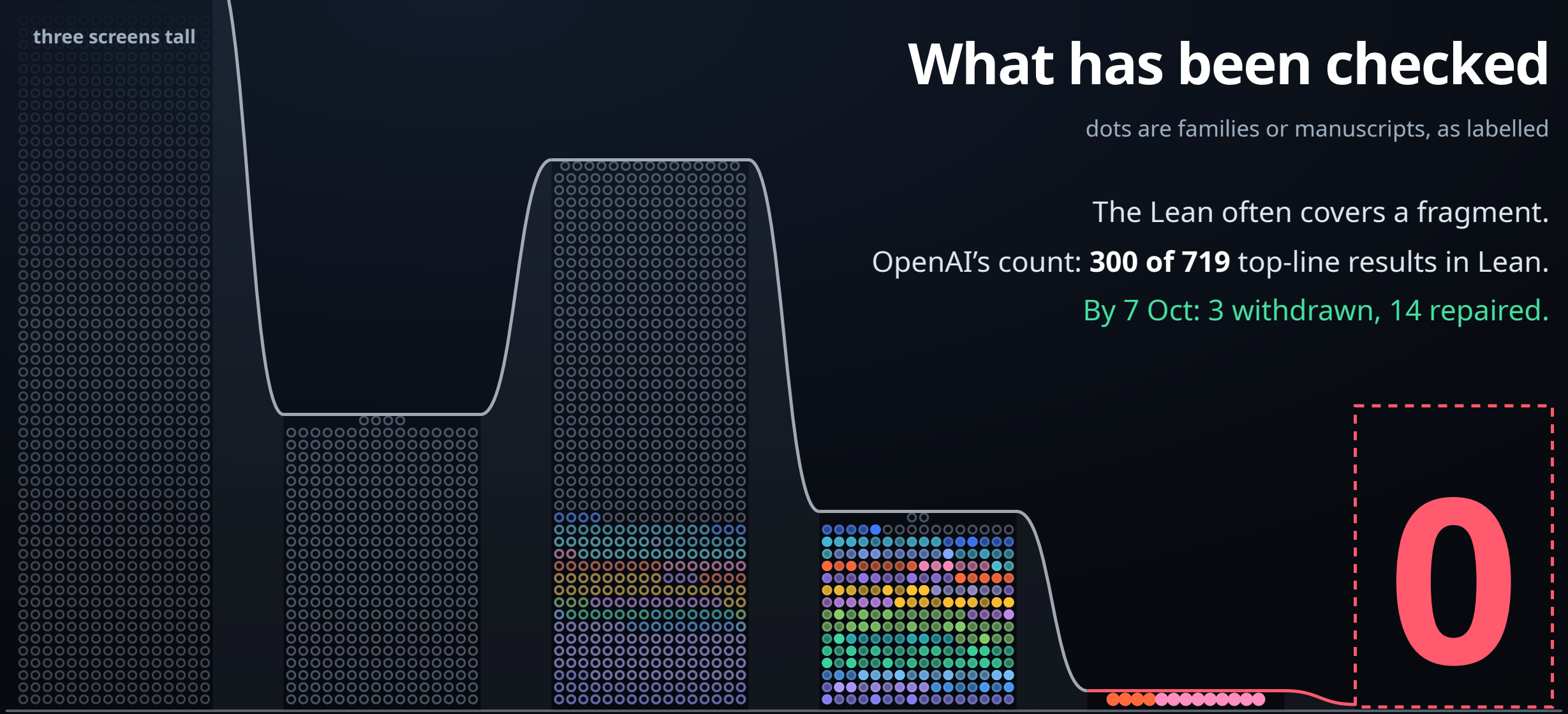
10 Sep
17 Sep
23–24 Sep
1 Oct
5–6 Oct

some Lean
no Lean
physics
everything else

github.com/openai/math at commit fd4aeeb (8 Oct 2026), cloned 12 Oct 2026 · folder dates, first edition · totals: README and a file scan

What has been checked

dots are families or manuscripts, as labelled



The Lean often covers a fragment.

OpenAI's count: **300 of 719** top-line results in Lean.

By 7 Oct: 3 withdrawn, 14 repaired.

~4,000
problems posed

372
result families

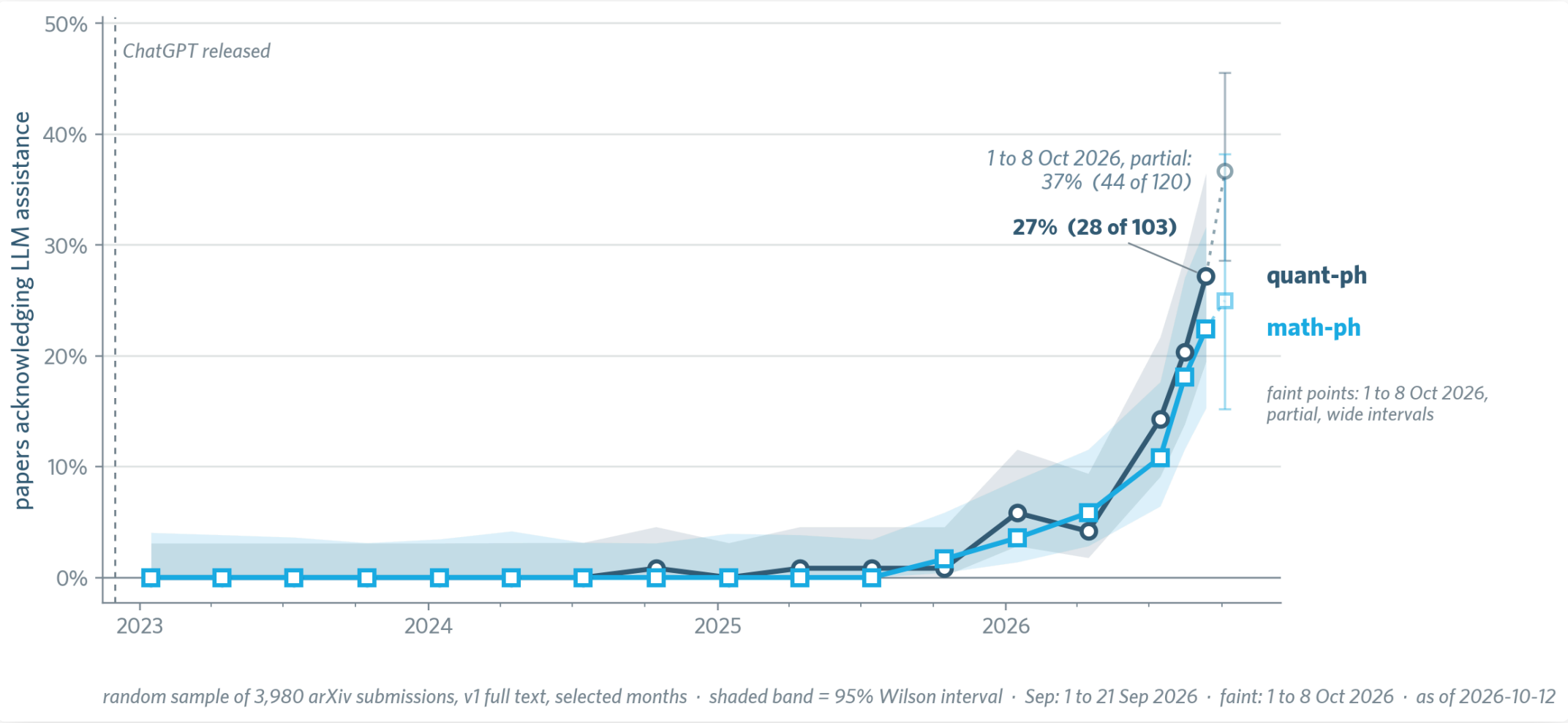
719
manuscripts

242
families with some Lean

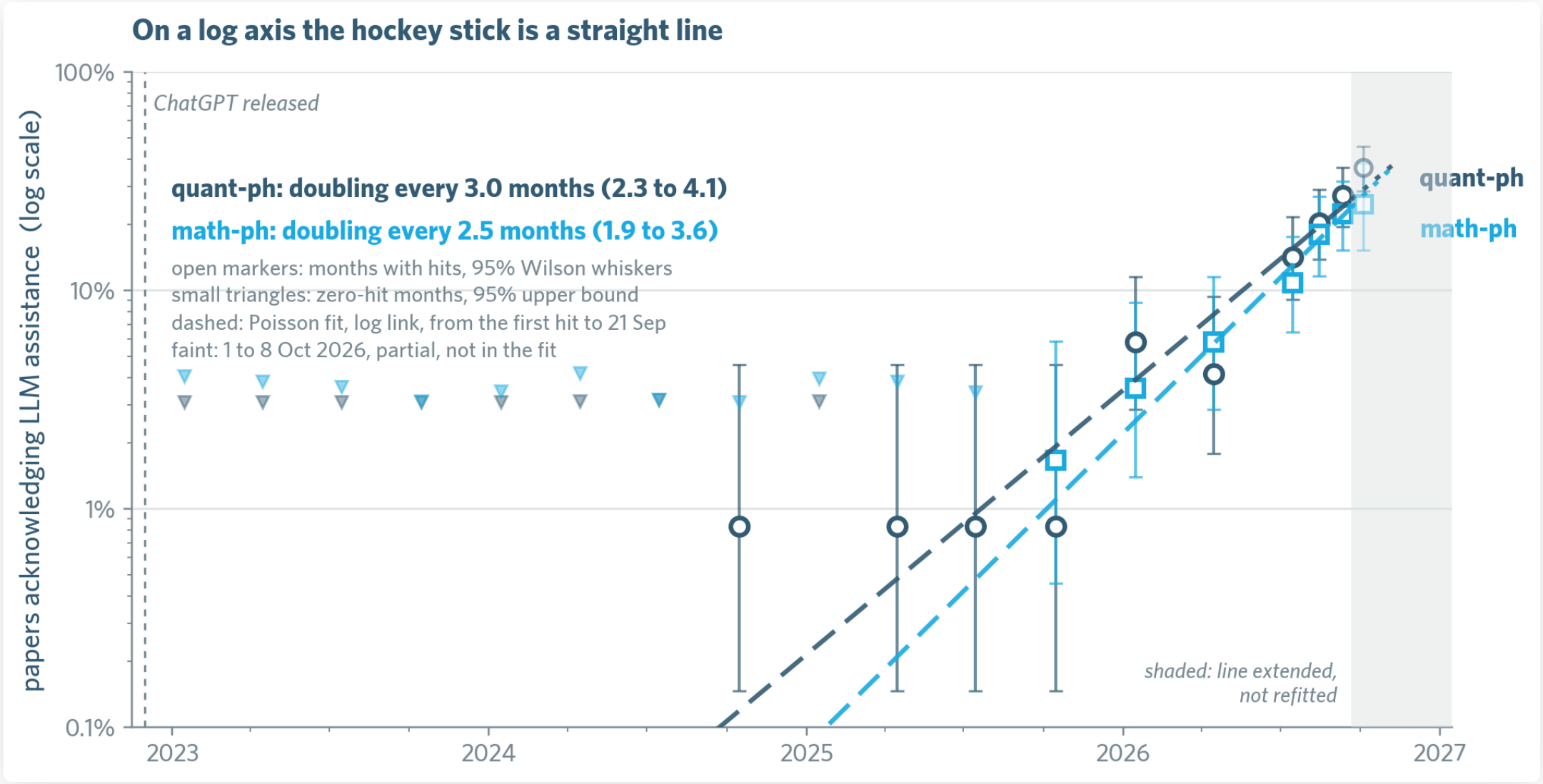
13 of 73
physics families with the headline in Lean

refereed
none found, 12 Oct

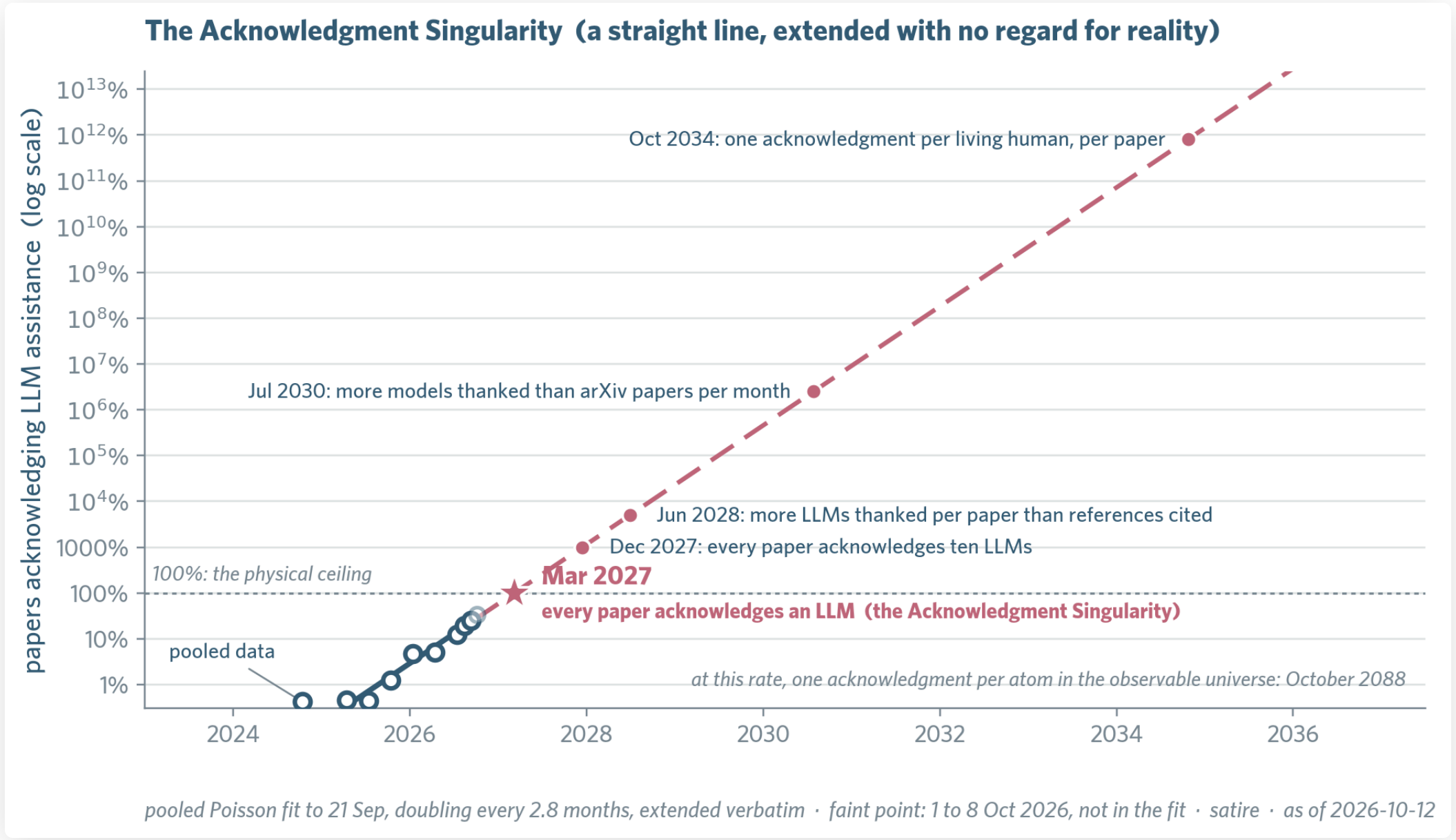
github.com/openai/math at commit fd4aeeb (8 Oct 2026), cloned 12 Oct 2026 · README, history.md, lean/docs · my count: physics blocks only



Papers whose acknowledgments name an LLM: quant-ph **0% through 2023, 27% in September 2026**, 37% in 1 to 8 October
3,980 sampled v1 submissions, every hit read by hand · a disclosure rate, not a usage rate · faint: 1 to 8 Oct, partial · 12 Oct 2026



Same data, log axis: **doubling every 2.8 months** since the first hit (95% CI 2.3 to 3.5)
fit to 21 Sep, extended: 28% predicted for 1 to 8 Oct · observed 33% (95% CI 27 to 40) · consistent with the line, no bend yet · 12 Oct 2026


The Acknowledgment Singularity (a straight line, extended with no regard for reality)
papers acknowledging LLM assistance (log scale)
10^13%
10^12%
10^11%
10^10%
10^9%
10^8%
10^7%
10^6%
10^5%
10^4%
1000%
100%
10%
1%
Oct 2034: one acknowledgment per living human, per paper
Jul 2030: more models thanked than arXiv papers per month
Jun 2028: more LLMs thanked per paper than references cited
Dec 2027: every paper acknowledges ten LLMs
100%: the physical ceiling
Mar 2027
every paper acknowledges an LLM (the Acknowledgment Singularity)
pooled data
at this rate, one acknowledgment per atom in the observable universe: October 2088
2024
2026
2028
2030
2032
2034
2036
pooled Poisson fit to 21 Sep, doubling every 2.8 months, extended verbatim · faint point: 1 to 8 Oct 2026, not in the fit · satire · as of 2026-10-12

The lag collapsed: from three years to ten months

SOFTWARE ENGINEERING

Jun 2021
Copilot: a machine writes plausible code

Mar 2024
Devin hyped, then debunked

Feb 2025
Claude Code: the tooling loop closes

Q2 2025
Engineers: it is good at my own job

Nov 2025
Opus 4.5 first past 80% SWE-bench

37 months | 19 months | 11 months | 12 months | 10 months



MATHEMATICS & THEORETICAL PHYSICS

Jul 2024
IMO silver: AlphaProof

Oct 2025
Ten Erdős problems: claimed, retracted

Jan 2026
Erdős #728: model → Lean → human

May 2026
Unit distance falls; Gowers would take it for the Annals

Sep 2026
GPT-6 Astra ships; Navier-Stokes in Lean

6 Oct 2026
openai/math: 719 manuscripts

You are here

Sydney, October 2026

Déjà vu

	SOFTWARE ENGINEERING, 2026	MATHEMATICS, SEP TO OCT 2026
Understanding	“What would it take for you to be comfortable shipping code without reading and understanding it?” Gergely Orosz, LDX3 New York	“[…] there is no one to ask when we get stuck in a proof.” Bryna Kra, on openai/math, 8 Oct 2026
Review	“Everyone is playing the theater of doing reviews.” An engineer, quoted by Orosz	“[…] the current volume is drowning the reviewers.” Bryna Kra, 13 Sep 2026
Debt	“Everything is broken.” Orosz, on software reliability	“Math slop causes technical debt in mathematics […]” Henry Cohn, 15 Sep 2026

Known failures, known fixes

~~finite context · no continual learning~~

Files as memory

~~hallucination~~

Ground truth on disk

~~nonzero error rate~~

Decompose and verify

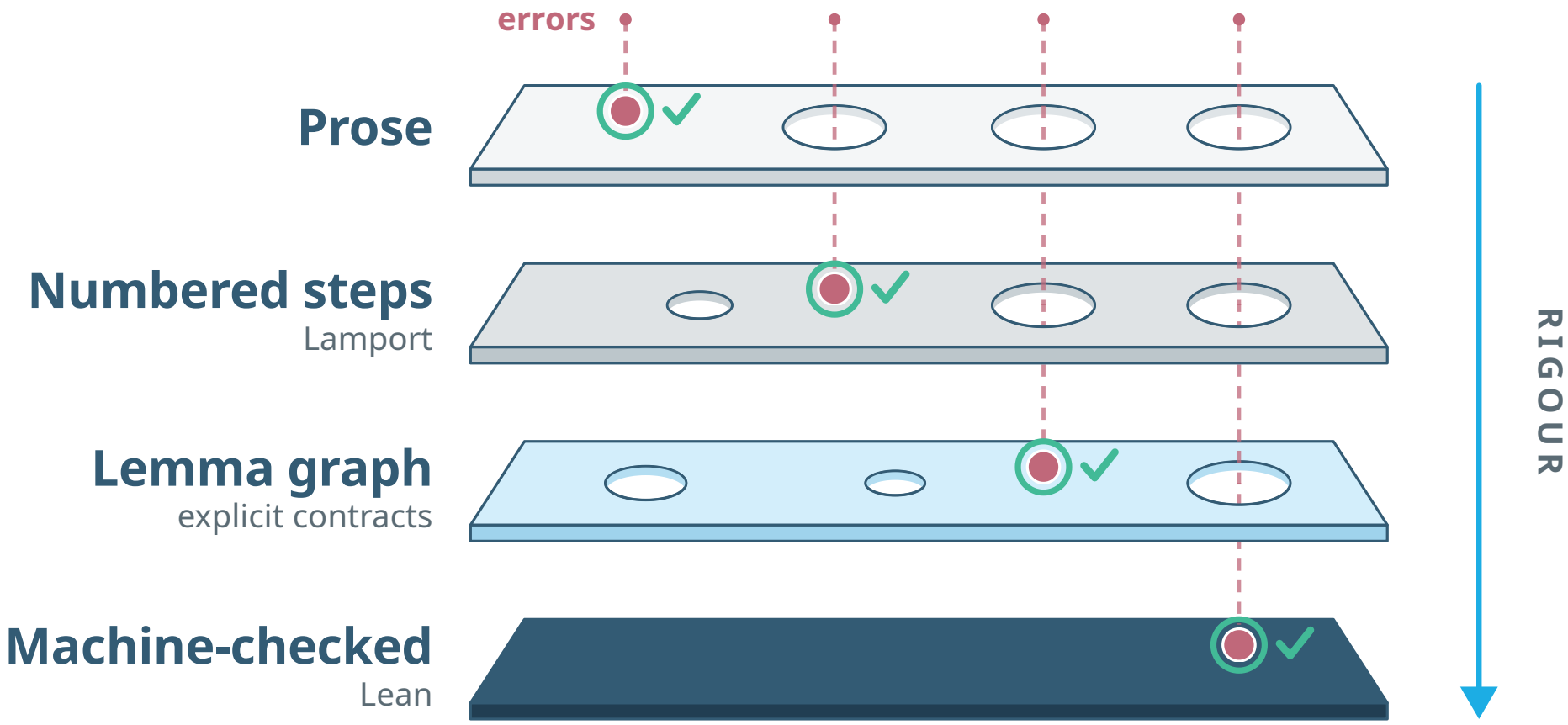
~~hyperfixation · sycophancy~~

Fresh adversarial check

Reliable computation out of unreliable components

Defence in depth

Swiss cheese model (Reason, 2000)



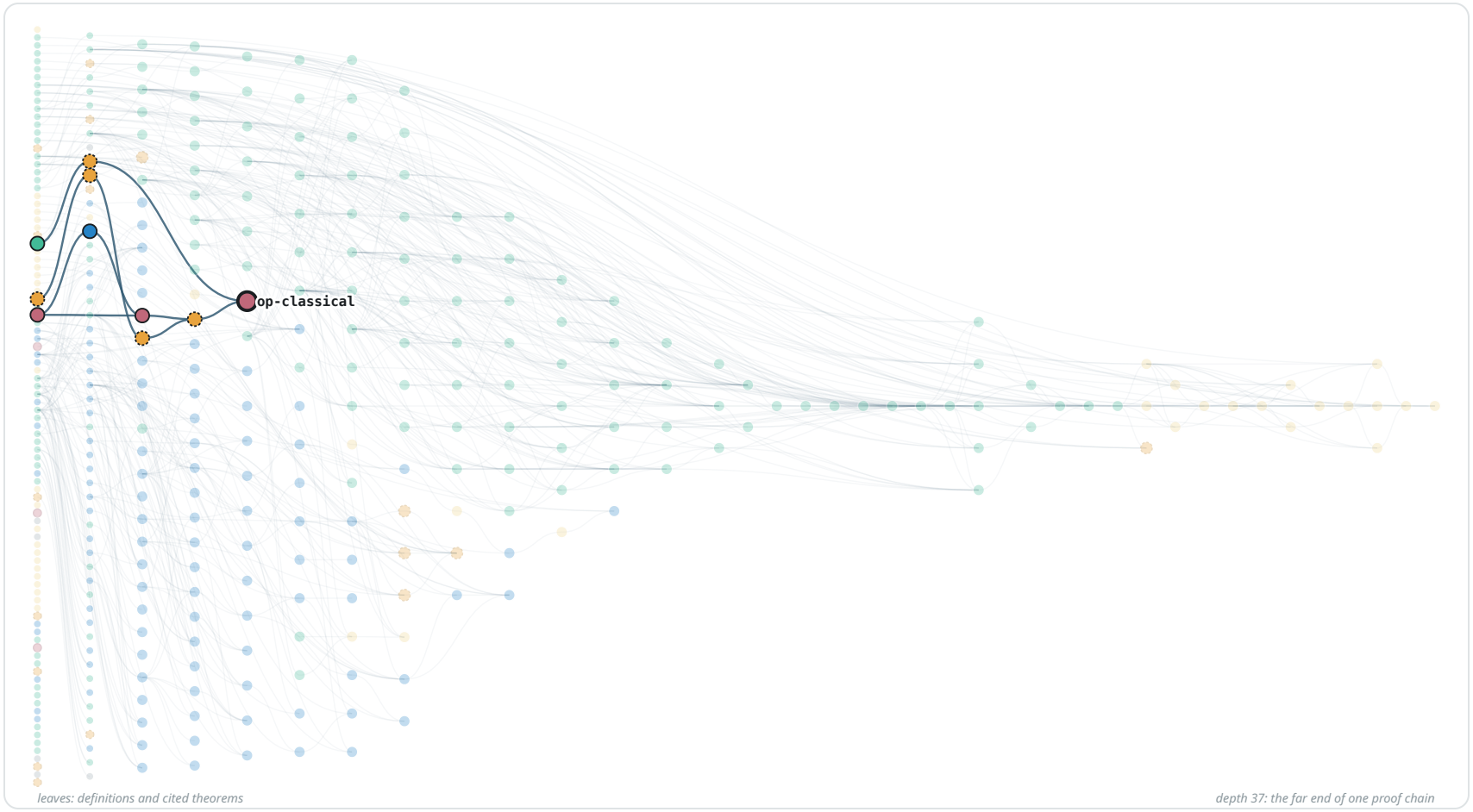
Prover → **Verifier** → **Formaliser** **adversarial agents push each claim one layer deeper**

schematic

github.com/tobiasosborne/vibefeld

What the Swiss cheese produces

one open problem, worked by agents for five weeks
364 results · 769 dependency edges · every arrow is a checked contract



click any node: what it rests on lights up · depth grows to the right

machine-validated: proved by one fresh agent, attacked by another	169
proved in prose, not yet adversarially checked	113
inherited paper-proof, unaudited here	22
conjecture, stated, heuristic or numerical	48
open problem	6
disproved or obstruction	6

op-classical

OPEN-PROBLEM OPEN

(OPEN) Classical projection stability: there are universal eta_0,C>0 (n-free) such that every row-stochastic Q with ||Q^2-Q||_{inf->inf} <= eta <= eta_0 admits a stochast...

rests on **9** results: **1** machine-validated · **1** prose · **7** unverified or open

The open problem itself. What it still rests on is orange and red: the gap is visible, named, and exactly where the next agent gets sent.

One agent proves, a fresh one attacks, a linker keeps the ledger. The colour is what survived.

Agents are amazing at helping you understand

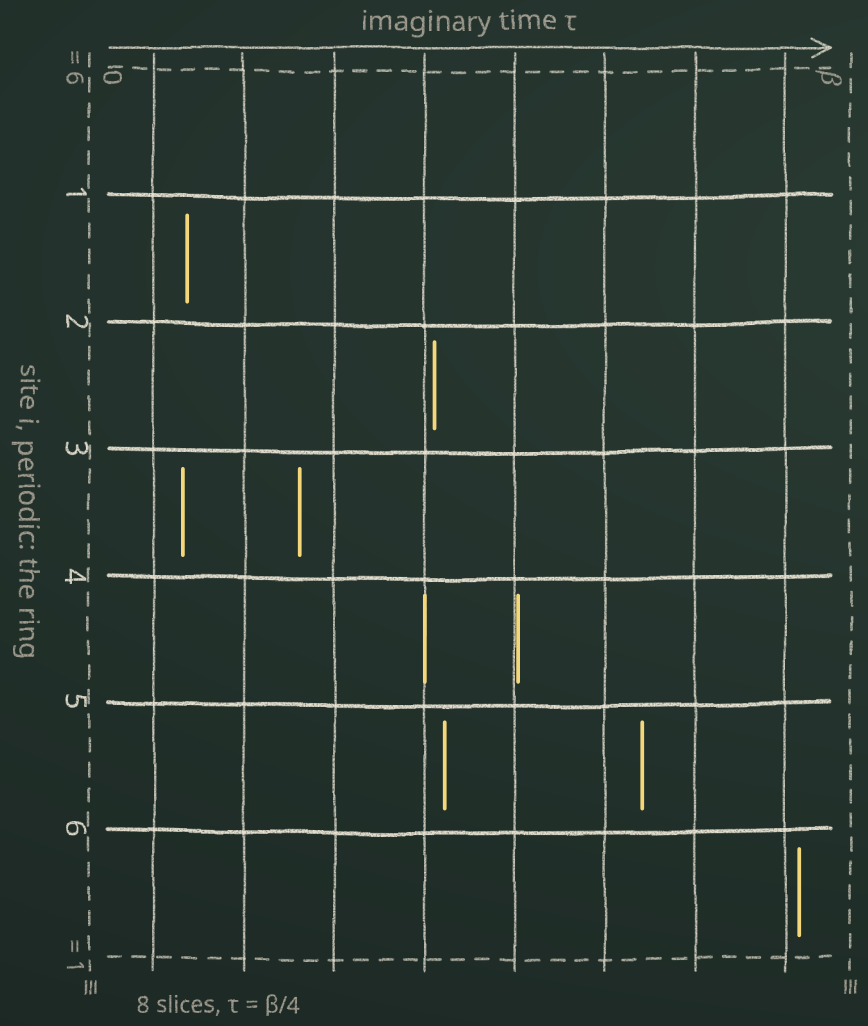
One of the 719: family 268, a proof that the spin-1 Heisenberg ring has a Haldane gap

READ	One agent refereed it from the LaTeX in a single pass and re-ran the paper's exact-arithmetic checks.
BUILD	Three agents, in parallel, each turned the paper, the referee report, an explainer and a one-page brief into a five-slide interactive explainer.
TIME	All three were back within 40 minutes. I picked one.

One partition function, two readings

$Z_n(\beta) = Tr\, e^{-\beta H_n}$, $H_n = \Sigma\, S_i S_{i+1}$ on a ring

toy: spin-½ ring, n = 6, β = 2 (keys [ ] change β, M changes the slices)



rotated: the operator now reads across bonds, one column at a time

Six worldlines, one per site. Height β. Bond events arrive as a Poisson process, rate 1 per bond per unit β: $e^{-\beta H} = e^{-\beta n/4}\, e^{\beta \Sigma Q_b}$, expanded in time-ordered singlet projectors Q_b. (Aizenman–Nachtergaele; the ticks are an illustration, the numbers below are not.)

READ ALONG TIME

Gibbs weights over 64 energy levels: $Z = \Sigma_k\, e^{-\beta E_k}$ = **657.1559026** (exact diagonalisation, uncut ring)

Cut β into 2M = 8 slices (toy lattice): $Z^{lat} = Tr[(T_A T_B)^M]$, 64 × 64 matrices = **681.135416929577**

READ ALONG SPACE

Transfer across one bond, acting on a column of 8 spins (256 × 256): $Z^{lat} = Tr[(X_e X_o)^{n/2}]$ = **681.135416929576**

Same number to 15 digits. Finer slices, M = 1, 2, 3, 4: `1136.9`, `758.17`, `700.36`, `681.14` → `657.16` (uncut).

One number, two probability vectors: $p_k \propto e^{-\beta E_k}$ over levels, $q_i \propto |\lambda_i|^n$ over the transfer spectrum.

toy model: spin-½ Heisenberg ring, 6 sites, computed live · the paper's kernel is the continuous-time one, real symmetric, spin 1 · openai/math, family 268, read 7 Oct 2026

The fine print

what is proved, what was checked, where a bug would hide

THEOREM 1.1

$H_L = \Sigma_j S_j S_{j+1}$, spin 1, periodic. Every even $L \geq 60$: unique ground state and $\gamma_L >$ **$(4/105)\log(80/79) \approx 4.8\cdot10^{-4}$**. Every even $L \geq 2304$: $\gamma_L >$ **$\log 20 / 784 \approx 3.8\cdot10^{-3}$**. Hence liminf $\gamma_L > 0$.

SCOPE

Even periodic rings and the liminf of their gaps. Cor 6.1 gives a gap inequality for subsequential limits of periodic ground states, not uniqueness or a gap for every infinite-volume ground state. True gap ≈ 0.41: the constants are a thousand times smaller.

THE SEEDS

Where the labour is. (60, 105/4) from exact Z_n, $n \leq 10$, at $\beta = 21/2$ (59,049 states, exact integers, rigorous series truncation), polynomial filters on the spectrum, a bond-dimension-26 MPS giving $Z_{120} > 1$, a 5/2-power raise of β. Margin: $S(60,105/4) > 7/8$ by $6\cdot10^{-4}$. Second seed (36, 49/4): defects 0.048 and 0.18, six doublings to (2304, 784).

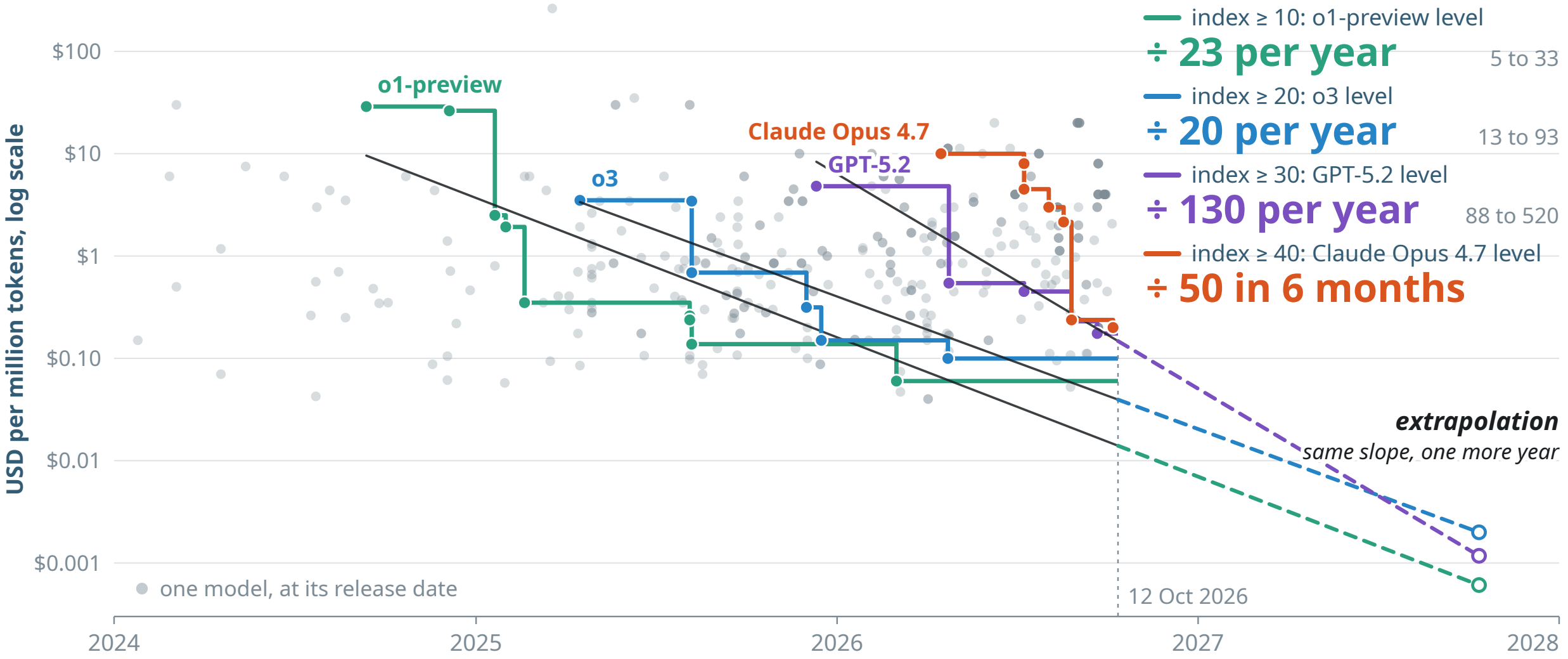
VERIFICATION

No Lean. Exact-rational scalar scripts and both MPS contractions re-run and pass; tightest margins $4.7\cdot10^{-5}$ and $6\cdot10^{-4}$. Independent exact diagonalisation reproduces the tables for $n \leq 9$. Unverified: $Z_{10}(21/2)$, which the $L \geq 60$ theorem needs, and $n = 10$ to 12 at $\beta = 49/4$.

VERDICT

One reviewer, one pass: no mathematical error found. **85 to 90%** that the theorem is proved as written, conditional on the $n = 10$ to 12 integer totals. If a bug hides anywhere, it is in the seeds, not in the doubling.

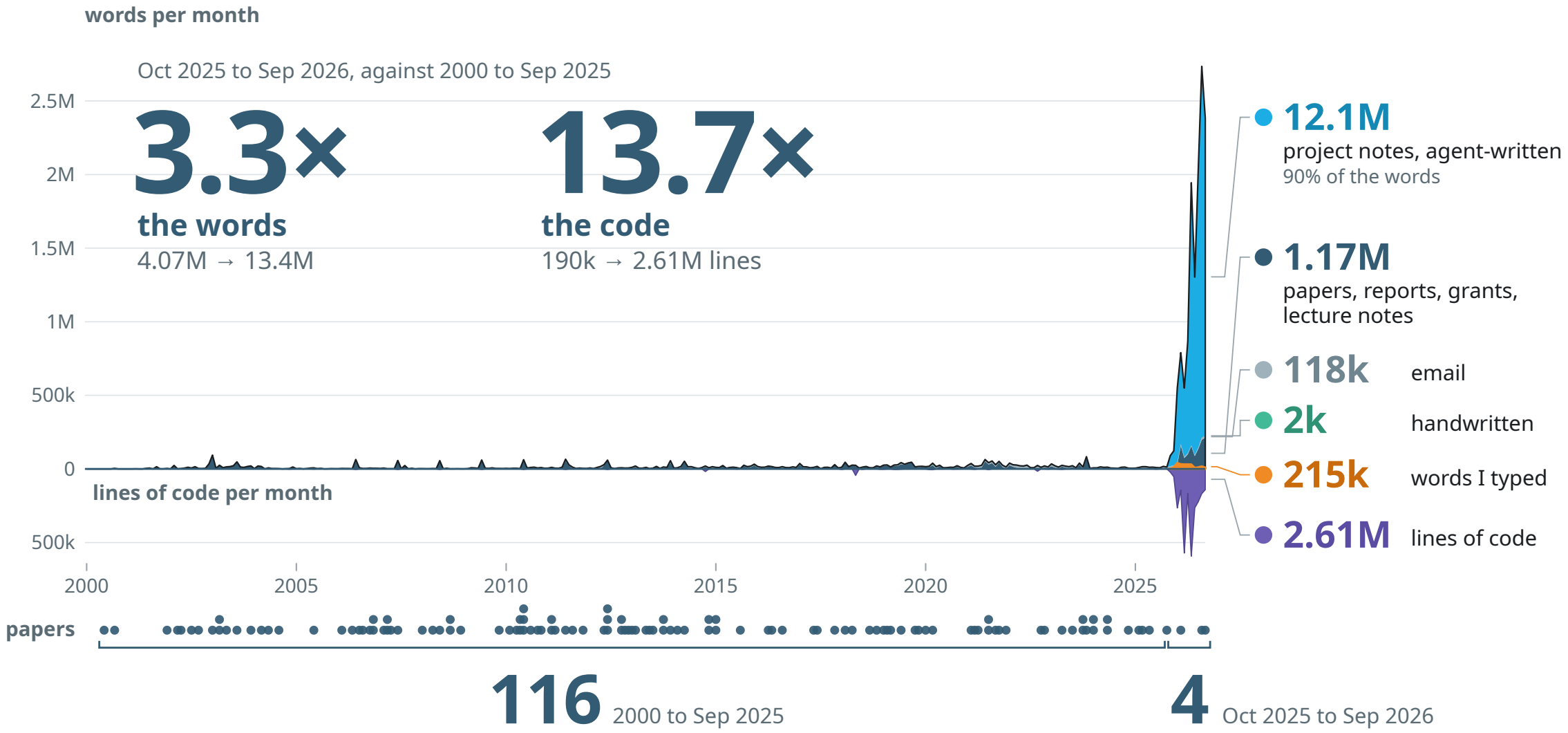
Price at fixed capability falls exponentially



Cheapest model at or above each level of the Artificial Analysis Intelligence Index.

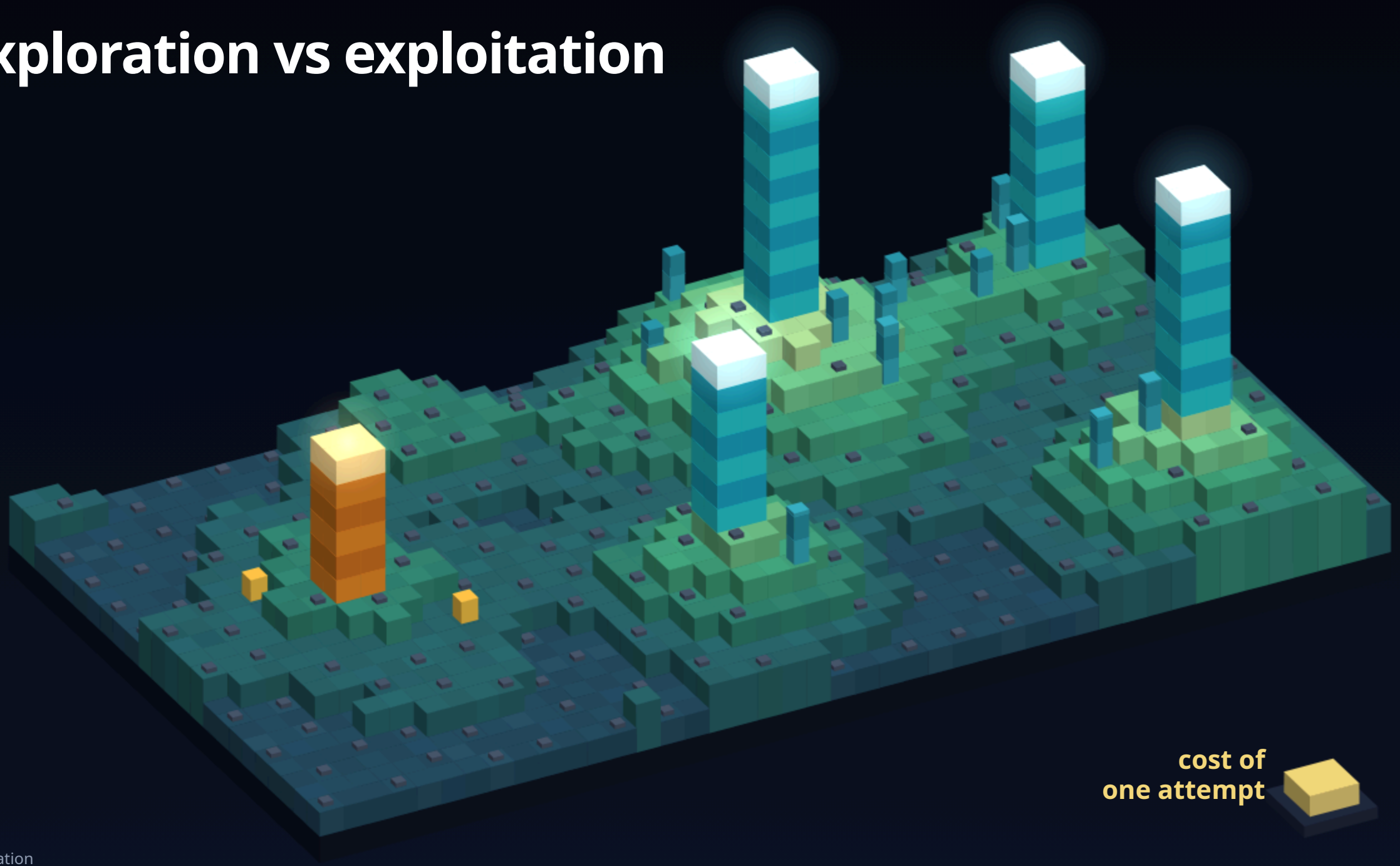
Artificial Analysis, Index v4.3.2, snapshot 2026-10-12. Price: list, blended 3 input to 1 output. Fits and 90% ranges: T. J. Osborne

I measured myself



Last twelve months: 90% of the words are agent-written project notes. Duplicates, drafts, third-party text removed. Papers: INSPIRE. Method: RESULTS.md, 27 Sep 2026.

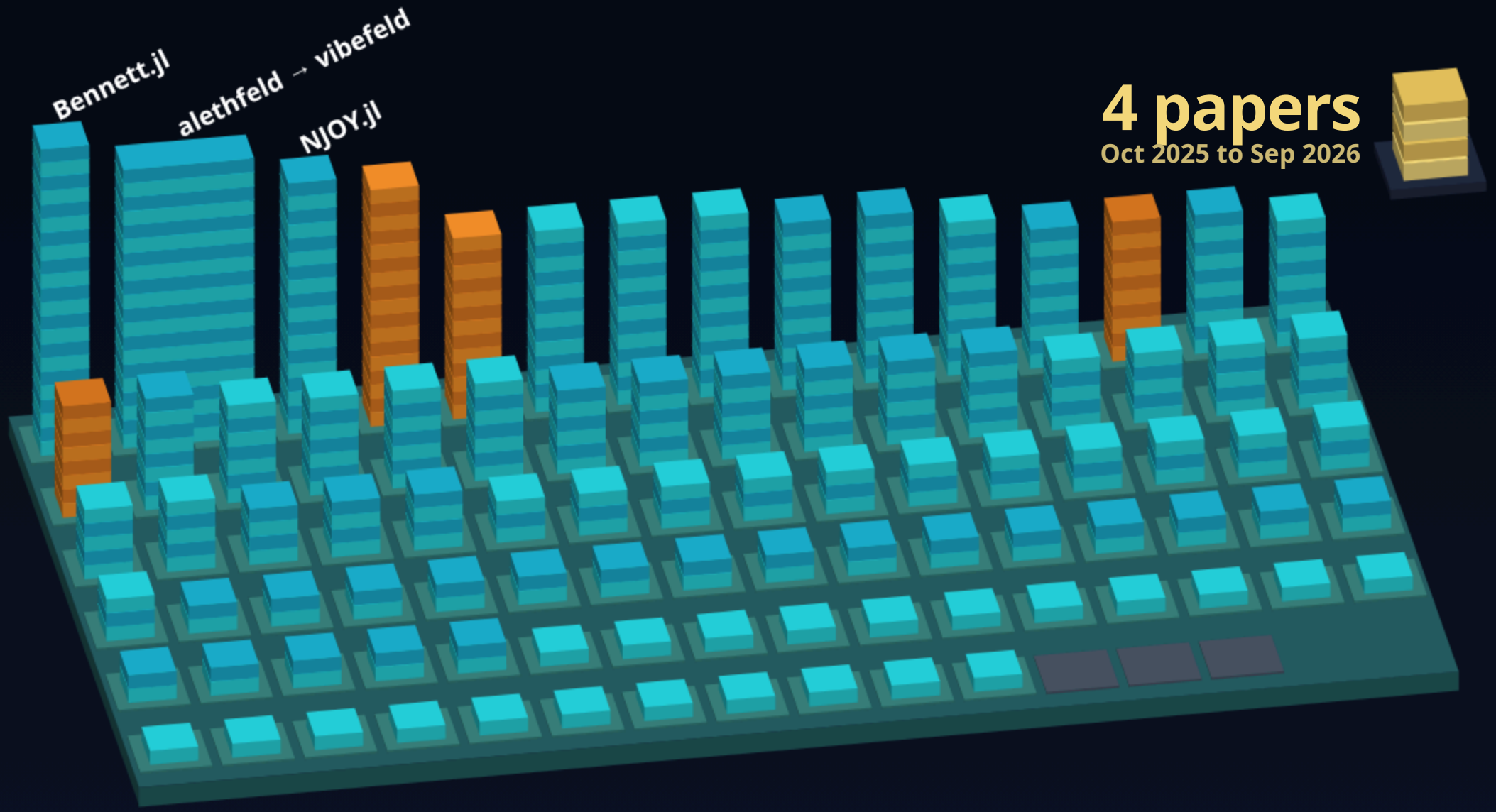
Exploration vs exploitation



illustration

94 repositories

one block: one week with commits
started 2014 to 2015



github.com/tobiasosborne, 13 Oct 2026 · grey plots: 3 forks · papers: INSPIRE

Output ≠ mission

THE WORK NOW

Effort

compute, years

Output

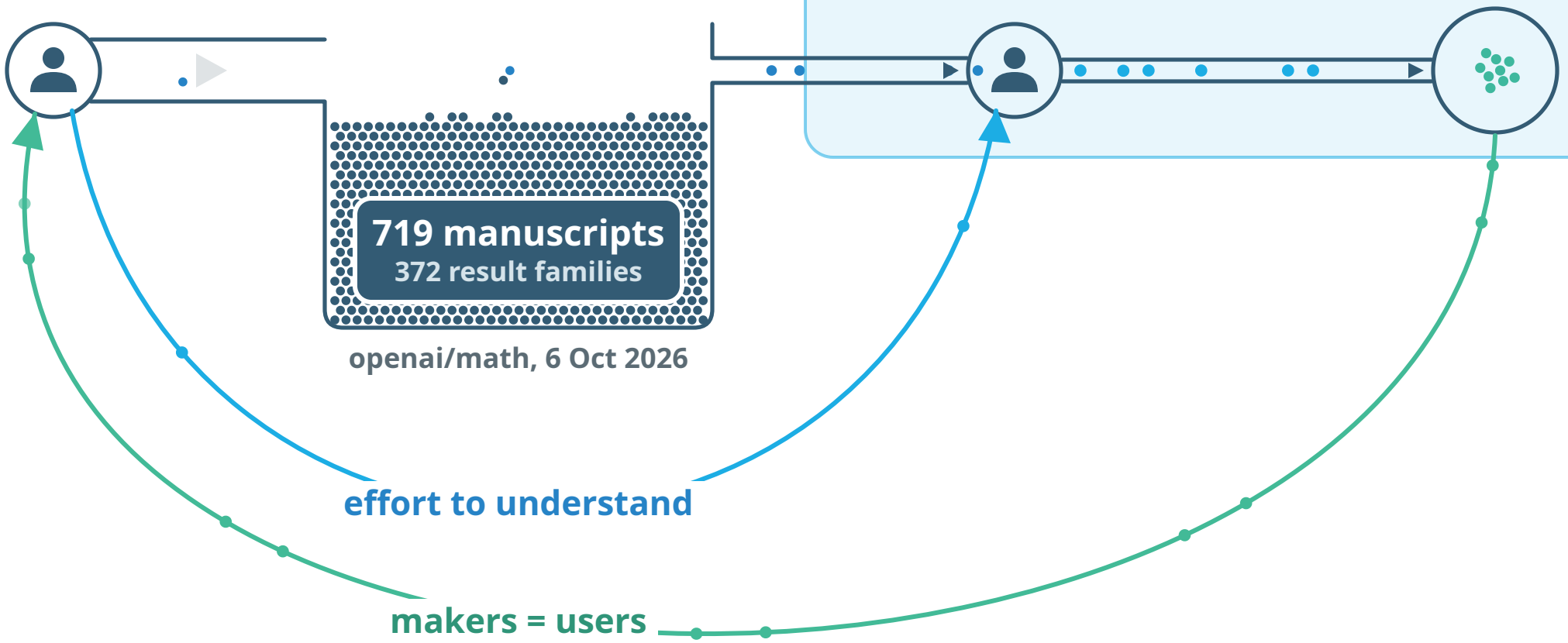
theorems, Lean certificates, problems closed

Outcome

mathematicians think, teach and ask differently

Mission

human understanding, passed on



adapted from Kent Beck

What works: four habits from software

01

“Ask the agent to explain things.”

“[…] you have a genius in your computer that can tell you exactly how it works.”

Theo Browne

A proof is no different.

02

Try five approaches

Being wrong is now cheap.

03

Read where a mistake matters

Statements, definitions, what others build on.

04

Make failures cheap to catch

Exact arithmetic, Lean, a second agent as referee.

“Use AI to get better.” Gergely Orosz

After Browne and Cramer (Convex Abstract panel, Sep 2026) and Orosz (LDX3 New York, 2026) · grey lines: my translation

You can outsource cognition
but you can't outsource *understanding*



The last arrow runs through us.

And we get to be here at the start.

How to train your slop cannon (Loader, Oppenheim, Osborne, 2026)

github.com/Open-Science-Ledger/how-to-train-your-slop-cannon

… when a new object comes within our reach,

the addition to our knowledge is the greater,

the more we already know;

so that it increases not as the new objects increase,

but in a much higher proportion. * * *



Colin Maclaurin
1698 to 1746

C. Maclaurin, *An Account of Sir Isaac Newton's Philosophical Discoveries* (1748), p. 392: the printed text ends here · drawing: Earl of Buchan, about 1795